

LABORATORY OBSERVATION

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 1
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Date: 29-08-2026

1. TITLE

Develop a College Information Portal Using HTML and HTML5 Features

2. AIM

To understand and implement the fundamental building blocks of HTML and HTML5 for creating a structured and interactive web page containing text, lists, tables, images, forms, semantic elements, multimedia, and HTML5 input types.

3. OBJECTIVE

- To understand the basic structure of an HTML5 document.
- To create web pages using headings and paragraphs.
- To apply different text-formatting tags.
- To create ordered and unordered lists.
- To add hyperlinks to external websites.
- To create and display student details using tables.
- To insert images with alternate text.
- To design a student registration form.
- To use HTML5 semantic elements such as header, nav, section, article and footer.
- To embed audio and video files.
- To implement HTML5 input types such as email, tel, date, color, number and range.
- To understand the role of HTML in structuring web content.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. HTML uses elements and tags to define different parts of a web page such as headings, paragraphs, lists, tables, forms, images, and links. HTML5 is the modern version of HTML that provides additional features such as semantic elements, multimedia support, and new form input types.

The basic structure of an HTML5 document consists of `<!DOCTYPE html>` which defines the document as an HTML5 document, `<html>` as the root element, `<head>` which contains metadata and the page title, and `<body>` which contains the visible content of the web page.

Headings and Paragraphs: `<h1>` to `<h6>` are used for headings, while `<p>` is used for paragraphs.

Text Formatting: `<b>` makes text bold, `<i>` makes text italic, `<u>` underlines text, `<sup>` displays superscript text, and `<sub>` displays subscript text.

Lists: creates an unordered list, while creates an ordered list. Individual items are created using .

Hyperlinks: The <a> tag is used to create links to other web pages or websites.

Tables: <table>, <tr>, <th>, and <td> are used to create and organize tabular information.

Images: The tag is used to display images. The alt attribute provides alternative text if the image cannot be displayed.

Forms: HTML forms collect user information using controls such as text boxes, radio buttons, checkboxes, dropdowns, textareas, and buttons.

Semantic Elements: HTML5 provides meaningful elements such as <header>, <nav>, <section>, <article>, and <footer> to describe the structure of a web page.

Multimedia: The <audio> and <video> elements allow audio and video files to be embedded directly into web pages.

HTML5 Input Types: HTML5 provides specialized input types such as email, tel, date, number, color, and range, which provide better input handling and browser-level validation.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and define it as an HTML5 document using <!DOCTYPE html>.
2. Create the basic <html>, <head>, and <body> structure and add the title College InformationPortal.
3. Add a <header> with the college name and a <nav> bar linking to each section.
4. Add headings and paragraphs containing college information.
5. Apply bold, italic, underline, superscript, and subscript formatting.
6. Create an ordered list of courses offered and an unordered list of college facilities.
7. Add hyperlinks to the college website and Google, opening in a new tab.
8. Create a student details table containing Roll No, Name, Branch, and Email.
9. Insert the college logo and campus/building image using the tag with alternate text.
10. Create a student registration form with name, email, password, phone number, date of birth, gender, hobbies, branch, favourite colour, age, programming skill (range) and address controls, plus submit and reset buttons.
11. Add HTML5 semantic elements such as header, navigation, section, article, and footer.
12. Embed an audio file using the <audio> element and a video file using the <video> element.
13. Save the HTML file, open it in a web browser, and verify all elements, links, images, formcontrols, audio, and video.
14. Record the output using screenshots.

6. PROGRAM

index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>

  <!-- Header -->
  <header>
    <h1 align="center">ABC College Information Portal</h1>
    <hr>
  </header>

  <!-- Navigation -->
  <nav align="center">
    <a href="#about">About</a> |
    <a href="#courses">Courses</a> |
    <a href="#students">Students</a> |
    <a href="#register">Registration</a> |
    <a href="#media">Media</a>
  </nav>

  <hr>

  <!-- About College -->
  <section id="about">
    <h2>About Our College</h2>

```

```

    <p>
      <b>ABC College of Engineering</b> is one of the best engineering colleges.
      It provides <i>quality education</i> with
      <u>excellent infrastructure</u> and experienced faculty members.
    </p>

    <p>
      Example of Subscript: H<sub>2</sub>O <br>
      Example of Superscript: x<sup>2</sup>
    </p>
  </section>

  <hr>

  <!-- Ordered List -->
  <section id="courses">
    <h2>Courses Offered</h2>

    <ol>
      <li>Computer Science Engineering</li>
      <li>Electronics and Communication Engineering</li>
      <li>Electrical Engineering</li>
      <li>Civil Engineering</li>
      <li>Mechanical Engineering</li>
    </ol>

    <h2>College Facilities</h2>

    <ul>

```

```
<h2>Useful Links</h2>

<a href="https://www.google.com" target="_blank">Google</a><br>
<a href="https://www.andhrauniversity.edu.in" target="_blank">
  College Website
</a>
```

```
</section>
```

```
<hr>
```

```
<!-- Table -->
```

```
<section id="students">
```

```
  <h2>Student Information</h2>
```

```
  <table border="1" cellpadding="8">
```

```
    <tr>
```

```
      <th>Roll No</th>
```

```
      <th>Name</th>
```

```
      <th>Branch</th>
```

```
      <th>Email</th>
```

```
    </tr>
```

```
    <tr>
```

```
      <td>101</td>
```

```
      <td>Anusha</td>
```

```
      <td>CSE</td>
```

```
      <td>anusha@gmail.com</td>
```

```
    </tr>
```

```
    <tr>
```

```
      <td>102</td>
```

```
      <td>Rahul</td>
```

```
      <td>ECE</td>
```

```
      <td>rahul@gmail.com</td>
```

```
    </tr>
```

```
    <tr>
```

```
      <td>103</td>
```

```
      <td>Priya</td>
```

```
      <td>EEE</td>
```

```
      <td>priya@gmail.com</td>
```

```
    </tr>
```

```
  </table>
```

```
</section>
```

```
<hr>
```

```
<!-- Images -->
```

```
<section>
```

```
  <h2>College Logo</h2>
```

```


<h2>College Building</h2>



</section>

<hr>

<!-- Form -->

<section id="register">

  <h2>Student Registration Form</h2>

  <form>

    <label>Name:</label><br>
    <input type="text" placeholder="Enter Name">
    <br><br>

    <label>Email:</label><br>
    <input type="email" placeholder="Enter Email">
    <br><br>

    <label>Password:</label><br>
    <input type="password">
    <br><br>

    <label>Phone Number:</label><br>
    <input type="tel">
    <br><br>

    <label>Date of Birth:</label><br>
    <input type="date">
    <br><br>

    <label>Gender:</label><br>

    <input type="radio" name="gender">Male

    <input type="radio" name="gender">Female

    <br><br>

    <label>Hobbies:</label><br>

    <input type="checkbox">Reading

    <input type="checkbox">Sports

    <input type="checkbox">Music

    <br><br>
```

```
<label>Branch:</label><br>

<select>

    <option>Select</option>
    <option>CSE</option>
    <option>ECE</option>
    <option>EEE</option>
    <option>Civil</option>
    <option>Mechanical</option>

</select>

<br><br>

<label>Favorite Color:</label><br>
<input type="color">

<br><br>

<label>Age:</label><br>
<input type="number">

<br><br>

<label>Programming Skill:</label><br>
<input type="range">

<br><br>
```

```
    <label>Favorite Color:</label><br>
    <input type="color">

    <br><br>

    <label>Age:</label><br>
    <input type="number">

    <br><br>

    <label>Programming Skill:</label><br>
    <input type="range">

    <br><br>

    <label>Address:</label><br>

    <textarea rows="5" cols="40"></textarea>

    <br><br>

    <input type="submit" value="Submit">

    <input type="reset" value="Reset">

</form>

</section>
```

```
<hr>

<!-- HTML5 Audio Video -->

<section id="media">

    <article>

        <h2>College Anthem (Audio)</h2>

        <audio controls>

            <source src="college.mp3" type="audio/mpeg">

            Your browser does not support audio.

        </audio>

    </article>

    <br>

    <article>

        <h2>College Introduction Video</h2>

        <video width="500" controls>

            <source src="college.mp4" type="video/mp4">

            <source src="college.mp4" type="video/mp4">

            Your browser does not support video.

        </video>

    </article>

</section>

<hr>

<!-- Footer -->

<footer align="center">

    <p>
        © 2026 ABC College Information Portal |
        Designed using HTML5
    </p>

</footer>

</body>
</html>
```

7. CODE EXPLANATION

Code / Tag	Explanation
<code><!DOCTYPE html></code>	Declares the document as HTML5.
<code><html></code>	Defines the root element of the web page.
<code><head></code>	Contains metadata and page information.
<code><title></code>	Specifies the title displayed in the browser tab.
<code><header></code>	Defines the header section of the page.
<code><nav></code>	Defines the navigation links section.
<code><h1>, <h2></code>	Define different levels of headings.
<code><p></code>	Defines a paragraph.
<code>, <i>, <u></code>	Display text in bold, italic and underline respectively.
<code><sup>, <sub></code>	Display superscript and subscript text.
<code>, , </code>	Create ordered and unordered lists and their items.
<code><a href target="_blank"></code>	Creates a hyperlink that opens in a new browser tab.
<code><table>, <tr>, <th>, <td></code>	Create and organize tabular student information.
<code></code>	Displays an image with alternate text if it cannot be loaded.
<code><form></code>	Creates the student registration form.
<code><input type="text/email/password/te l/date/radio/checkbox/color/number /range"></code>	Different HTML5 form controls, each with browser-level validation/UI.

<code><select>, <option></code>	Creates the Branch dropdown list.
<code><textarea></code>	Creates the multi-line Address input area.
<code><input type="submit"/"reset"></code>	Submit and reset buttons for the form.
<code><section id="..."></code>	Groups related content into named sections.
<code><article></code>	Defines independent, self-contained media content (audio/video blocks).
<code><audio controls><source></audio></code>	Embeds the college anthem audio file with playback controls.
<code><video controls><source></video></code>	Embeds the college introduction video file with playback controls.
<code><footer></code>	Defines the footer section with the copyright notice.

8. TEST CASES AND EXECUTION DATA

Test Case	Test / Input	Expected Result	Actual Result	Status
TC-01	Open index.html	College Information Portal should be displayed	Portal displayed correctly	Pass
TC-02	Check headings and paragraphs	Headings and college information should be displayed correctly	Displayed correctly	Pass
TC-03	Check text formatting	Bold, italic, underline, superscript and subscript should appear	Formatting displayed correctly	Pass
TC-04	Check ordered and unordered lists	Both lists should be displayed correctly	Lists displayed correctly	Pass
TC-05	Click College Website / Google link	Links should open in a new browser tab	Links opened successfully	Pass
TC-06	Check student table	Student details should be displayed in tabular form	Table displayed correctly	Pass
TC-07	Check logo and building image	Images should attempt to load with alternate text on failure	Alt text displayed for missing images	Pass
TC-08	Enter details in registration form	User should be able to enter/select all field types	All form controls worked correctly	Pass
TC-09	Check HTML5 input types	Email, tel, date, color, number and range controls should be available	Input types displayed correctly	Pass
TC-10	Check audio and video	Audio and video controls should be displayed	Multimedia controls displayed successfully	Pass
TC-11	Check footer	Footer should appear at the bottom with the copyright text	Footer displayed correctly	Pass

9. OUTPUT

Output 1: College Information Portal

ABC College Information Portal

[About](#) | [Courses](#) | [Students](#) | [Registration](#) | [Media](#)

About Our College

ABC College of Engineering is one of the best engineering colleges. It provides *quality education* with excellent infrastructure and experienced faculty members.

Example of Subscript: H₂O

Example of Superscript: x²

Courses Offered

1. Computer Science Engineering
2. Electronics and Communication Engineering
3. Electrical Engineering
4. Civil Engineering
5. Mechanical Engineering

College Facilities

- Library
- Computer Labs
- Sports Complex
- Hostel
- Cafeteria

Useful Links

[Google](#)
[College Website](#)

Student Information

Roll No	Name	Branch	Email
101	Anusha	CSE	anusha@gmail.com
102	Rahul	ECE	rahul@gmail.com
103	Priya	EEE	priya@gmail.com

College Logo

 College Logo

College Building

 College Building

Student Registration Form

Name:

Enter Name

Email:

Enter Email

Password:

Phone Number:

Date of Birth:

mm / dd / yyyy

Gender:

☐ Male

☐ Female

Hobbies:

☐ Reading

☐ Sports

☐ Music

Branch:

Select

Favorite Color:

Age:

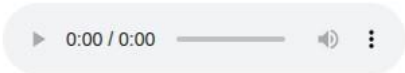
Programming Skill:

Address:

Submit

Reset

College Anthem (Audio)



College Introduction Video



TASK 2 — HTML5 CANVAS AND CSS STYLING

1. TITLE

Implement HTML5 Canvas Graphics and Design a Styled Web Page Using CSS

2. AIM

To understand and implement HTML5 Canvas for drawing basic graphical elements and internal CSS3 for styling and designing a structured, Flexbox-based web page.

3. OBJECTIVE

- To understand the HTML5 <canvas> element.
- To obtain the 2D drawing context using JavaScript.
- To draw basic shapes using Canvas: a rectangle, circle, and line.
- To display text using Canvas.
- To understand internal CSS styling.
- To apply fonts, colors, gradients, spacing, borders, and shadows.
- To create a navigation bar using CSS Flexbox.
- To design card-based content using Flexbox.
- To apply hover effects to links, cards and buttons.
- To create a visually styled web page.

4. THEORY

HTML5 Canvas: The HTML5 Canvas element provides a drawing area that can be controlled using JavaScript. It can be used to create graphics, shapes, text, animations, and other visual elements. The <canvas> element defines the drawing area, while JavaScript is used to perform the actual drawing. The getContext("2d") method obtains the 2D drawing context, which provides methods for drawing shapes and text.

In this task: fillRect() is used to draw a rectangle, arc() is used to create a circular path, fill() fills the circular path, moveTo()/lineTo() define a line and stroke() draws it, fillText() is used to display text, fillStyle sets the fill colour, and font defines the text font.

CSS: CSS (Cascading Style Sheets) is used to control the appearance and layout of HTML elements. In this task, CSS is written internally inside a <style> block and is used for font styling, background gradient, navigation styling with Flexbox, a Flexbox-based card layout, borders, rounded corners, box shadows, button styling, hover effects, and footer styling.

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document and add an internal <style> block inside <head>.
2. Add a heading inside the <header>.
3. Create a navigation bar containing Home, About, Services, and Contact links using an unorderedlist arranged with Flexbox.
4. Add a <canvas> element with a width of 700 pixels and height of 300 pixels, and assign it an ID.

5. Use JavaScript to access the canvas element and obtain the 2D drawing context using `getContext("2d")`.
6. Set the fill colour and draw a rectangle using `fillRect()`.
7. Create a circular path using `arc()` and fill it using `fill()`.
8. Create a line using `moveTo()` and `lineTo()`, set its width and colour, and draw it using `stroke()`.
9. Set the font and colour, and display text using `fillText()`.
10. Create a container for the content cards and add separate cards for HTML5, CSS3, and JavaScript, each with a heading, description, and a Read More button, arranged using Flexbox.
11. Create a footer.
12. Apply a gradient background to the page and style the header, navigation links, cards and buttons.
13. Apply borders, rounded corners, and shadows to the cards, and hover effects to navigation links, cards and buttons.
14. Open the HTML file in a browser and verify the canvas graphics and the styled layout.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Styled Webpage</title>

<style>
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}

body{
    font-family:Arial, sans-serif;
    background:linear-gradient(to right,#74ebd5,#ACB6E5);
}

/* Header */
header{
    background:#003366;
    color:white;
    text-align:center;
    padding:25px;
    text-shadow:2px 2px 5px black;
}
```

```
nav{
  background:#222;
  padding:15px;
}

nav ul{
  list-style:none;
  display:flex;
  justify-content:center;
}

nav ul li{
  margin:0 20px;
}

nav ul li a{
  color:white;
  text-decoration:none;
  font-size:18px;
  padding:8px 15px;
  border-radius:5px;
  transition:0.3s;
}

nav ul li a:hover{
  background:orange;
  color:black;
}
```

```
.canvas-section{
  text-align:center;
  padding:30px;
}

canvas{
  border:3px solid black;
  background:white;
  box-shadow:5px 5px 10px gray;
}

/* Cards */
.cards{
  display:flex;
  justify-content:center;
  gap:20px;
  flex-wrap:wrap;
  padding:30px;
}

.card{
  width:250px;
  background:white;
  padding:20px;
  border:2px solid gray;
  border-radius:15px;
  box-shadow:5px 5px 10px gray;
  text-align:center;
  transition:0.3s;
```

```
.card:hover{
    transform:translateY(-8px);
}
```

```
.card h2{
    color:darkblue;
    margin-bottom:10px;
}
```

```
.card p{
    margin-bottom:15px;
}
```

```
button{
    background:blue;
    color:white;
    border:none;
    padding:10px 20px;
    border-radius:10px;
    cursor:pointer;
    transition:0.3s;
}
```

```
button:hover{
    background:green;
}
```

```
/* Footer */
```

```
footer{
    background:#111;
    color:white;
    text-align:center;
    padding:20px;
    margin-top:20px;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>My Styled Webpage</h1>
```

```
<p>HTML5 Graphics and CSS3 Styling</p>
```

```
</header>
```

```
<nav>
```

```
<ul>
```

```
<li><a href="#">Home</a></li>
```

```
<li><a href="#">About</a></li>
```

```
<li><a href="#">Services</a></li>
```

```
<li><a href="#">Contact</a></li>
```

```
</ul>
```

```
</nav>
```

```
<section class="canvas-section">
```

```
<h2>HTML5 Canvas Example</h2>
```

```
<canvas id="myCanvas" width="700" height="300"></canvas>
```

```
</section>
```

```

<section class="cards">

<div class="card">
  <h2>HTML5</h2>
  <p>HTML5 is the latest version of HTML used to build webpages.</p>
  <button>Read More</button>
</div>

<div class="card">
  <h2>CSS3</h2>
  <p>CSS3 styles webpages using colors, fonts, borders and layouts.</p>
  <button>Read More</button>
</div>

<div class="card">
  <h2>JavaScript</h2>
  <p>JavaScript adds interactivity and dynamic behavior to websites.</p>
  <button>Read More</button>
</div>

</section>

<footer>
  <p>© 2026 My Styled Webpage | Designed by Govindamma</p>
</footer>

<script>

```

```

let canvas=document.getElementById("myCanvas");
let ctx=canvas.getContext("2d");

// Rectangle
ctx.fillStyle="skyblue";
ctx.fillRect(30,30,150,80);

// Circle
ctx.beginPath();
ctx.arc(280,70,40,0,2*Math.PI);
ctx.fillStyle="red";
ctx.fill();

// Line
ctx.beginPath();
ctx.moveTo(380,30);
ctx.lineTo(650,130);
ctx.lineWidth=5;
ctx.strokeStyle="green";
ctx.stroke();

// Text
ctx.font="28px Arial";
ctx.fillStyle="blue";
ctx.fillText("Govindamma",180,220);

</script>

</body>

```

7. CODE EXPLANATION

Code / Tag	Explanation
<code><canvas></code>	Creates a drawing area on the web page.
<code>id="myCanvas"</code>	Provides an identifier for accessing the canvas using JavaScript.
<code>width="700", height="300"</code>	Sets the canvas width and height in pixels.
<code>getElementById()</code>	Retrieves the canvas element from the HTML document.
<code>getContext("2d")</code>	Obtains the 2D drawing context.
<code>fillStyle</code>	Specifies the fill colour of graphical elements.
<code>fillRect()</code>	Draws and fills a rectangle.
<code>beginPath()</code>	Starts a new drawing path.
<code>arc()</code>	Creates an arc or circular path.
<code>fill()</code>	Fills the current path.
<code>moveTo()</code>	Sets the starting point of a line.
<code>lineTo()</code>	Specifies the ending point of a line.
<code>lineWidth, strokeStyle, stroke()</code>	Set the line width/colour and draw the current path.
<code>font, fillText()</code>	Specify the font and size, and display text on the canvas.
<code><style></code>	Contains all internal CSS rules for the page.
<code>linear-gradient()</code>	Creates the diagonal background gradient for the page.
<code><header>, text-shadow</code>	Defines the header section and adds a shadow effect to its text.
<code><nav>, , display:flex</code>	Creates a horizontal navigation bar arranged using Flexbox.
<code>:hover</code>	Applies styles when the mouse pointer moves over a link, card or button.

<code>.cards, display:flex, flex-wrap</code>	Arranges the HTML5/CSS3/JavaScript cards in a responsive, wrapping row.
<code>.card (border, border-radius, boxshadow)</code>	Styles each card with a border, rounded corners and a drop shadow.
<code>.card:hover, transform:translateY()</code>	Lifts a card slightly upward when hovered.
<code>button, button:hover</code>	Styles the Read More buttons and their hover colour change.
<code><footer></code>	Defines the footer section with the copyright notice.

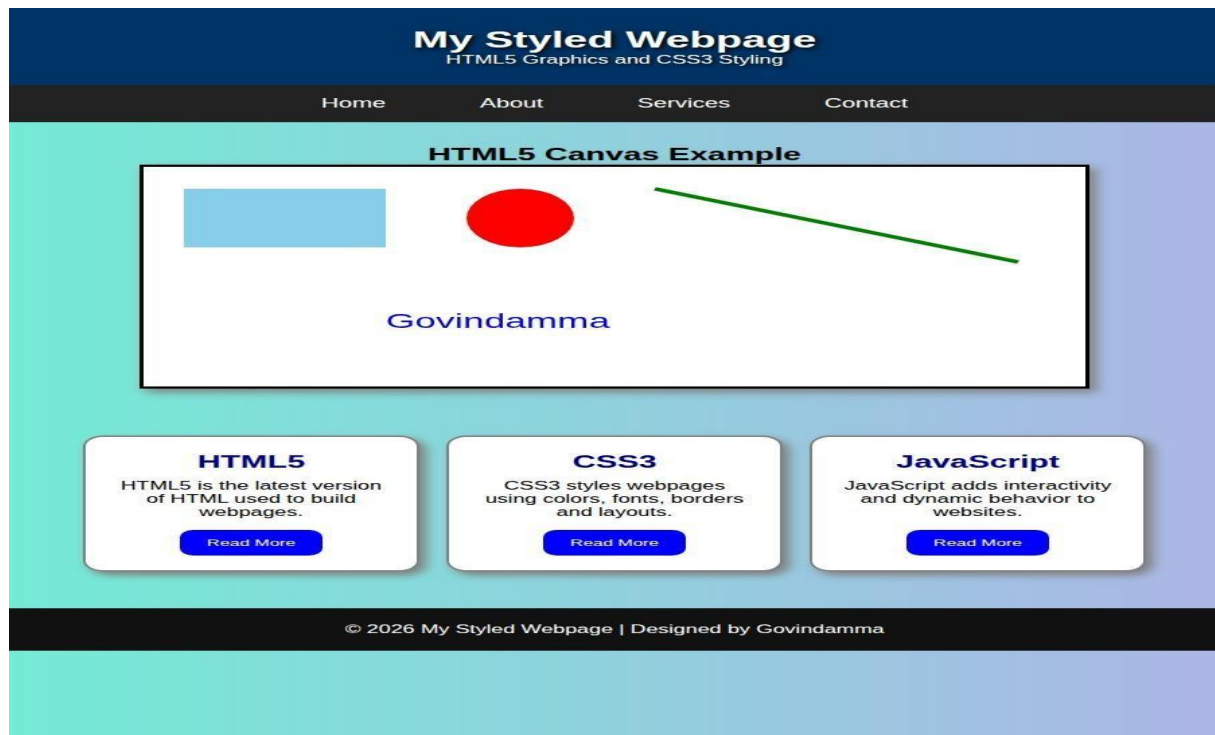
8. TEST CASES AND EXECUTION DATA

The following test cases were recorded after executing index.html in a browser.

Test Case	Test / Input	Expected Result	Actual Result	Status
TC-01	Open Canvas/CSS HTML page	Styled web page with canvas should be displayed	Displayed correctly	Pass
TC-02	Execute rectangle code	A filled rectangle should appear on the canvas	Rectangle displayed correctly	Pass
TC-03	Execute circle code	A filled circle should appear on the canvas	Circle displayed correctly	Pass
TC-04	Execute line code	A line should appear on the canvas	Line displayed correctly	Pass
TC-05	Execute text code	Text should appear on the canvas	Text displayed correctly	Pass
TC-06	Check header	Header should be centered and styled with a text shadow	Displayed correctly	Pass
TC-07	Check navigation links	Navigation links should be displayed horizontally using Flexbox	Displayed correctly	Pass
TC-08	Hover over navigation link	Hover background effect should appear	Hover effect displayed	Pass
TC-09	Check content cards	Three cards (HTML5, CSS3, JavaScript) should be displayed using Flexbox	Cards displayed correctly	Pass
TC-10	Check card styling	Cards should have borders, rounded corners and shadows	Styling applied correctly	Pass
TC-11	Hover over card / button	Card should lift slightly / button colour should change	Hover effects displayed	Pass
TC-12	Check footer	Footer should appear at the bottom with styled text	Footer displayed correctly	Pass

9. OUTPUT

Output: HTML5 Canvas and CSS Styled Web Page



10.RESULT

Thus, HTML5 Canvas graphics were successfully implemented using JavaScript to draw a rectangle, circle, line, and text. A styled web page was also successfully developed using internal CSS, including Flexbox-based navigation and card layout, borders, shadows, gradients, buttons, and hover effects. The outputs were verified successfully.